

Assignment :- 2 (CA-Network)

1. Parallel computing Example : Weather forecasting systems

Application :-

National weather service and organizations like NASA use supercomputers that run parallel computing models to simulate and predict weather patterns.

How it works :-

- The atmosphere is divided into 3D grids.
- Each grid is assigned to a processor or core, which simultaneously computes temperatures, wind, pressure, etc.
- These results are then combined to produce a global forecast.

Why parallel computing is important :-

- Weather data changes rapidly - real-time prediction requires massive computations.
- Parallel processing reduces runtime from days to minutes or hours.
- More cores = more accurate and faster predictions.

2. Networked Systems Example : Online multiplayer Gaming.

Application :-

Games like PUBG, Call of Duty, or Fortnite use client-server architectures with networked systems to enable real-time multiplayer gameplay.

How it works :-

- Game clients (player's devices) communicate with central game servers over the Internet.
- Servers handle logic like player position, movement, actions and environment updates.

- All devices are synchronized via a reliable, low latency network.

Why networked systems are important:-

- Enables real-time interaction among millions of players worldwide.
- Keeps gameplay consistent and fair.
- Supports cloud-based scaling (e.g., more servers during high traffic).